



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER II

2024/2025

SECP 2753 - DATA MINING

SECTION 1

LECTURER: DR. ROLIANA BINTI IBRAHIM

ASSIGNMENT 1: ANALYSIS OF RESEARCH DATA ABOUT LIVER DISEASE
AND ABALONE

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160

1.0 Introduction

Data mining is a process to break down large dataset to smaller dataset to identify patterns and relationships that are useful in decision-making [7]. This report will explore two real-world dataset: Abalone dataset and Liver Disease dataset provided by the instructor. Both dataset focuses on different thing where the Abalone dataset focuses on the prediction of the age of abalones using physical measurement, while Liver Disease dataset consists of biochemical data of a male individual in identifying early signs of liver damage or disease by considering the enzyme levels and alcohol consumption of a person. This report looks at both datasets to understand their basic details, explore useful patterns, and apply data processing techniques like normalization and discretization to the liver dataset. These steps help make the data easier to understand and more useful for analysis.

2.0 Dataset Overview

In this subtopic, we will describe the overview of both datasets and basic information about the data.

2.1 Abalone Dataset

Instances: 4,177 abalones

Features: 8 input variables + 1 target variable

Continuous: Length, Diameter, Height, Whole weight, Shucked weight, Viscera weight, Shell weight

Categorical: Sex (Male(M), Female(F), Infant(I))

Target: Determining the age of abalone by its rings (integer count; $\text{Age} \approx \text{Rings} + 1.5$ years)

Variables	Description	Data Type	Classification
Sex	Categorical: Male (M), Female (F), Infant (I)	Categorical	Nominal
Length	Longest shell measurement (mm)	Continuous	Ratio

Diameter	Perpendicular to length (mm)	Continuous	Ratio
Height	Height with meat in shell (mm)	Continuous	Ratio
Whole weight	Weight of whole abalone (grams)	Continuous	Ratio
Shucked weight	Weight of meat (grams)	Continuous	Ratio
Viscera weight	Gut weight after bleeding (grams)	Continuous	Ratio
Shell weight	After being dried (grams)	Continuous	Ratio
Rings	Count of growth rings	Integer	Discrete/Ratio

Table 1: Info About The Attributes in Abalone Dataset

2.2 Liver Disease Dataset

Instances: 345 individuals

Features: 6 input variables (excluding the unnamed first column)

Continuous:

- mcv (Mean corpuscular volume)
- alkphos (Alkaline phosphatase)
- sgpt (Alanine aminotransferase)
- sgot (Aspartate aminotransferase)
- gammagt (Gamma-glutamyl transpeptidase)
- drinks (Number of half-pint equivalents of alcoholic beverages drunk per day)

Categorical:

All features in this dataset are numerical, hence, there none explicitly present in this dataset.

Target:

The dataset does not explicitly provide the target, however, the drinks column can be used to represent a level of a drinker. For example, it can categorizing individuals as heavy drinkers vs. non-heavy drinkers

Additional Information:

- All instances are from male individuals.
- All dataset have numerical value with no missing value.
- The dataset shows no direct indicator of the diagnosis of liver disease.
- The unnamed first column is simply an index and can be ignored.

Below is the data description for the Liver Disease Dataset chosen represented in Table xxx below. The dataset

Variables	Description	Type of Variable	Measurement Level
mcv	Mean corpuscular volume	Quantitative	Ratio
alkphos	Alkaline phosphatase	Quantitative	Ratio
sgpt	Alanine aminotransferase	Quantitative	Ratio
sgot	Aspartate aminotransferase	Quantitative	Ratio
gammagt	Gamma-glutamyl transpeptidase	Quantitative	Ratio
drinks	Number of half-pint equivalents of alcoholic beverages drunk per day	Quantitative	Ratio

Table 2: Info About The Attributes in Liver Dataset

3.0 Dataset Insights

In this section, we are going to elaborate on the insight found from these datasets.

3.1 Abalone Dataset

Descriptive statistical analysis reveals that the mean length, diameter, and height are 0.524, 0.408, and 0.140 units, respectively. Weight measurements indicate that the average Whole Weight is 0.829 grams. Moving on to the abalone rings analysis, most abalones from the dataset have 8 to 11 rings, corresponding to an age range of approximately 9.5 to 12.5 years. For the Sex variable, males make up about 36.6% of the dataset, followed by infants at 32.1% and females at 31.3%. This relatively balanced distribution supports unbiased gender-based analyses, though the slight male predominance should be considered. Moving on to the correlation analysis, below is the table of correlation heatmap based on abalone dataset.

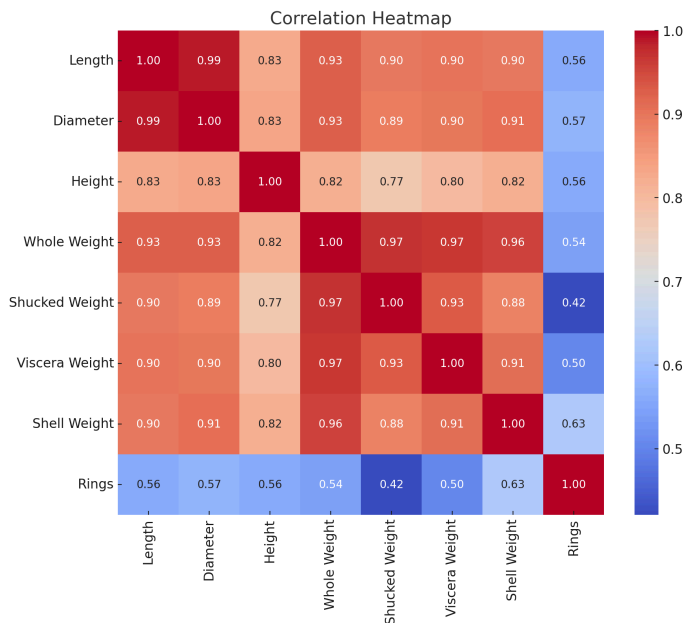


Figure 1: Correlation Heatmap of Abalone Dataset

Correlation analysis shows strong positive relationships among the continuous variables. Length and Diameter have a near-perfect correlation (0.99), indicating that as abalones grow longer, they also grow wider. Whole Weight and Shucked Weight are similarly highly correlated (0.97). Shell Weight shows the strongest correlation with Rings (0.63), making it a key predictor of age, while Length and Diameter have moderate correlations

(around 0.56–0.57). Shell Weight emerges as the most reliable age indicator, though not all attributes are equally predictive; for instance, Shucked Weight has a weaker link to age. Some anomalies, such as zero values for Height, suggest measurement errors that require data cleaning. Overall, the abalone dataset offers strong potential for predictive modeling based on physical characteristics.

3.2 Liver Disease Dataset

Descriptive statistical analysis of the liver disease dataset shows that the mean values for key liver biomarkers are as follows: alkphos– 69.87, sgpt – 30.41, sgot – 24.64, and gammagt – 38.28. High reading of these enzymes can be a signal for liver damage or disease [2]. For drinks attribute, it records the average alcohol beverages intake in a day and it has a mean of 3.46. Too much consumption of alcohol in a day can be a risk factor for liver disease where the dataset shows the maximum value for drinks attribute is 20[4]. The mcv attribute with mean value of 90.16 can be a useful indicator as macrocytosis can be a sign of liver disease[3]. Correlation analysis was conducted to explore the relationships between these biomarkers and alcohol consumption patterns as shown below in the correlation heatmap of liver disease dataset:

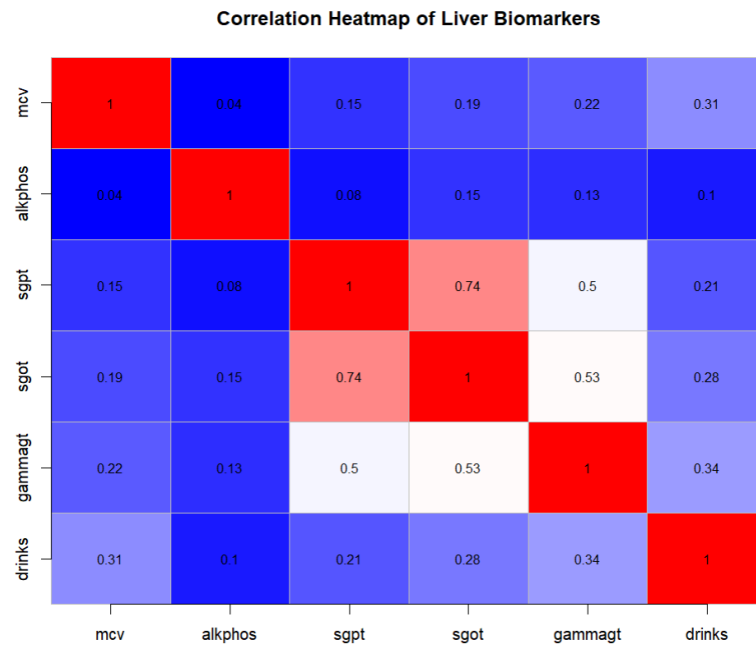


Figure 1: Correlation Heatmap of Liver Disease Dataset

The heatmap shows that some of the liver enzyme levels are closely related to each other. For example, sgpt and sgot have a strong positive correlation (0.74), which may point to a common underlying liver issue when both are elevated. sgot is also moderately linked to gammagt (0.53), and sgpt to gammagt (0.50), suggesting that these enzymes have tendencies to rise together when the liver is under stress. Among all, gammagt has the strongest link with alcohol intake (0.34), making it a useful marker for alcohol-related liver problems. There's also a moderate correlation between mcv and drinks (0.31), which fits with what we know about alcohol's effect on red blood cell size. Hence, these patterns suggest that early liver disease can be detected on enzyme readings and drinking habits of a person.

4.0 Data Processing Method

For this task, the liver dataset is chosen to go through data processing which will focus on normalization and discretization only.

4.1 Normalization

To ensure that all numerical values are within common scale, for the normalization process, we applied min-max normalization. This method rescales the values of an attribute into the range [0.0, 1.0] using the formula as shown below:

$$v' = \frac{v - \min(v)}{\max(v) - \min(v)} \times (\text{new_max} - \text{new_min}) + \text{new_min}$$

Where:

- v = original value
- v' = normalized value
- $\min(v)$ = minimum values of the attribute
- $\max(v)$ = maximum values of the attribute
- new_min = target range (0. 0)
- new_max = target range (1. 0)

The process was implemented using RStudio using base R. The code to the normalization process was as shown in Figure 3. The code also includes the first 5 rows of both original

dataset and normalized dataset for comparison for a clear insight by referring to Figure 4 and Figure 5. A full normalized dataset can be obtained in appendices.

```
# Load required libraries
library(gridExtra)
library(grid)

# Define columns to normalize
columns_to_normalize <- c('mcv', 'alkphos', 'sgpt', 'sgot', 'gammagt')

# Custom Min-Max normalization function
custom_min_max_norm <- function(x, new_min, new_max) {
  ((x - min(x)) / (max(x) - min(x))) * (1.0 - 0.0) + 0.0
}

# Copy dataset (replace 'liver' with your actual data frame if different)
liver_normalized <- liver

# Apply normalization
for (col in columns_to_normalize) {
  liver_normalized[[col]] <- custom_min_max_norm(liver[[col]], new_min = 0, new_max = 1)
}

# Get first 5 rows from both datasets for comparison
sample_original <- head(liver[columns_to_normalize], 5)
sample_normalized <- head(liver_normalized[columns_to_normalize], 5)

# Create table plots
table_before <- tableGrob(sample_original, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 10)),
  colhead = list(fg_params = list(fontface = "bold", fontsize = 12, col = "white"), bg_params = list(fill = "darkred"))
))

table_after <- tableGrob(sample_normalized, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontface = "bold", fontsize = 10)),
  colhead = list(fg_params = list(fontface = "bold", fontsize = 12, col = "white"), bg_params = list(fill = "darkgreen"))
))

# Add titles above each table
title_before <- textGrob("Before Normalization", gp = gpar(fontsize = 14, fontface = "bold"))
title_after <- textGrob("After Normalization", gp = gpar(fontsize = 14, fontface = "bold"))

# Combine titles and tables
before_combined <- gTree(children = gList(title_before, table_before), vp = viewport(layout.pos.row = 1))
after_combined <- gTree(children = gList(title_after, table_after), vp = viewport(layout.pos.row = 1))

# Arrange side by side
grid.newpage()
grid.arrange(
  arrangeGrob(title_before, table_before, ncol = 1),
  arrangeGrob(title_after, table_after, ncol = 1),
  ncol = 2
)

#summary of whole attribute
summary(liver_normalized[columns_to_normalize])

#whole dataset normalized
write.csv(liver_normalized, "liver_normalized.csv", row.names = FALSE)
```

Figure 3: R code for normalization process

mcv	alkphos	sgpt	sgot	gammagt
85	92	45	27	31
85	64	59	32	23
86	54	33	16	54
91	78	34	24	36
87	70	12	28	10

Figure 4: First 5 rows of original dataset (before normalization)

mcv	alkphos	sgpt	sgot	gammagt
0.526315789473684	0.6	0.271523178807947	0.285714285714286	0.089041095890411
0.526315789473684	0.356521739130435	0.364238410596026	0.350649350649351	0.0616438356164384
0.552631578947368	0.269565217391304	0.19205298013245	0.142857142857143	0.167808219178082
0.684210526315789	0.478260869565217	0.198675496688742	0.246753246753247	0.106164383561644
0.578947368421053	0.408695652173913	0.0529801324503311	0.298701298701299	0.0171232876712329

Figure 5: First 5 rows of normalized dataset (after normalization)

4.2 Discretization

For this process, we are using the binning method where we divide the range into N intervals of equal size. The formula is $W = (B-A)/N$ where W is the width interval, B is highest value, A is minimum value and N is the number of bins. The binning method will include all liver dataset attributes except index. All attributes will be inserted into three bins. First bin symbolizes Low, second bin is for Medium and third bin will be for High. The detailed process will be shown in the table below. Full discretized dataset can be obtained in appendices.

Attributes	mcv	alkphos	sgpt	sgot	gammagt	drinks
Minimum (A)	65	23	4	5	5	0
Maximum (B)	103	138	155	82	297	20
No of bin	3	3	3	3	3	3
Width of interval (W)	$(103-65)/3 = 12.67$	$(138-23)/3 = 38.33$	$(155-4) / 3 = 50.33$	$(82-3) / 3 = 26.33$	$(297-5) / 3 = 97.33$	$(20-0) / 3 = 6.67$

Table 3: Discretization Process in Liver Dataset

Attribute	Bin 1 (Low)	Bin 2 (Medium)	Bin 3 (High)
mcv	[65, 77.67]	[77.67, 90.34]	[90.84,103]
alkphos	[23, 61.33]	[61.33, 99.66]	[99.66,138]
sgpt	[4, 54.33]	[54.33, 104.66]	[104.66, 155]

sgot	[5, 30.67]	[30.67, 56.34]	[56.34, 82]
gammagt	[5, 102.33]	[102.33, 199.66]	[199.66, 297]
drinks	[0, 6.67]	[6.67, 13.34]	[13.34, 20]

Table 4: Bin Range for Liver Dataset

5.0 Discussion

Upon examining the distribution trends, it was observed that the majority of the drinks attribute values were concentrated within the "Low" bin, suggesting that most individuals in the sample reported low alcohol consumption. Meanwhile, attributes such as gammagt and alkphos (alkaline phosphatase) exhibited a wider spread across all three bins, implying variability in liver enzyme activity and indicating differing liver health statuses among participants. The individuals in the "High" bin of alkphos are more likely to have liver disease [5]. Further clinical insights emerged from the discretization of sgpt and sgot, both critical indicators of liver function. Individuals categorized within the "High" bins for these enzymes may be at greater risk of liver inflammation or damage, emphasizing the clinical value of such categorization for early detection of liver-related conditions [2].

6.0 Conclusion

In summary, in Task, we gain valuable insights from both the dataset as well as understand more about the topic studied in it. Moving on to Task 2, the application of normalization and discretization techniques on the liver dataset helped enhance the interpretability and readiness of the data for further analysis. Through binning, continuous variables such as liver enzyme levels and alcohol intake were transformed into categorical intervals, allowing for clearer insight into their distribution and clinical implications. This transformation highlighted patterns such as the predominance of low alcohol consumption and the variability of enzyme levels among individuals, which may be linked to differing stages of liver health. These preprocessing steps are crucial for effective data mining, especially in health-related datasets where early pattern detection can lead to better diagnostic decisions.

7.0 References

- [1] Nash, W., Sellers, T., Talbot, S., Cawthorn, A., & Ford, W. (1994). *Abalone [Dataset]*. UCI Machine Learning Repository. <https://doi.org/10.24432/C55C7W>.
- [2] Daniel, C. (n.d.). *What Liver Enzymes Do and What Levels Are Normal*. Verywell Health.
Retrieved April 29, 2025, from <https://www.verywellhealth.com/liver-enzymes-1759916>
- [3] *Macrocytosis: Causes, Symptoms & Treatment*. (2023, March 14). Cleveland Clinic.
Retrieved April 29, 2025, from
<https://my.clevelandclinic.org/health/diseases/24814-macrocytosis>
- [4] *Alcohol-related liver disease*. (2023, May 29). NHS inform. Retrieved April 29, 2025, from
<https://www.nhsinform.scot/illnesses-and-conditions/stomach-liver-and-gastrointestinal-tract/alcohol-related-liver-disease/>
- [5] *Why Your Alkaline Phosphatase Level Might Be Low*. (2023, September 9). Health. Retrieved April 29, 2025, from <https://www.health.com/alkaline-phosphatase-7556288>
- [6] Forsyth, R. S. (n.d.). *Liver Disorders*. UCI Machine Learning Repository. Retrieved April 29, 2025, from <https://doi.org/10.24432/C54G67>
- [7] Gillis, A. S. (n.d.). *What is data mining? | Definition from TechTarget*. TechTarget. Retrieved April 29, 2025, from
<https://www.techtarget.com/searchbusinessanalytics/definition/data-mining>

8.0 Appendices

Normalized Liver Disease Dataset

<https://drive.google.com/file/d/1A5MMVG8YJCSUy5FiFTyPclbS8lJjEAo9/view?usp=sharing>

Discretized Liver Disease Dataset

<https://drive.google.com/file/d/1D8wmhl3dAlqrm654oxUToFa4uNYmkcSo/view?usp=sharing>